

Hard Spectral Rank

1024

512

256

128

64

32

$$\Delta_{1,2} = \beta_{\text{soft}} - \beta_{\text{hard}}$$

AdamW +0.37

Dion (1/2) +0.13

Dion (1/16) +0.24

Muon +0.14

NorMuon +0.21

768

1536

3072

6144

FFN Hidden Dimension (D)

- AdamW: $\beta = 0.29$
- Dion (1/2): $\beta = 0.79$
- Dion (1/16): $\beta = 0.50$
- Muon: $\beta = 0.82$
- NorMuon: $\beta = 0.80$